Alvaro Perdomo Gutierrez

Senior DevOps Engineer

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PROFESSIONAL SUMMARY

Senior DevOps Engineer with **12+** years across healthcare, fintech, and gaming infrastructure, building secure, scalable cloud systems on Azure and AWS with Infrastructure as Code. Experienced in containerization, CI/CD automation, and compliance-driven environments using Terraform, Docker, Kubernetes, and GitHub Actions. Ready to drive infrastructure reliability and security for enterprise healthcare platforms.

SKILLS

Languages: Python, Go, Bash, JavaScript, TypeScript, SQL, HTML, CSS

Cloud & DevOps: Azure, Terraform, Docker, Kubernetes, GitHub Actions, Azure Container Apps, Azure Key Vault, Network Security Groups, API Gateway, Cloudflare, AWS EC2, AWS S3, AWS Lambda, CloudWatch, Jenkins, GitLab CI, ArgoCD, Helm, Linux, Apache

Database: PostgreSQL, Redis, MongoDB, TcaplusDB, Elasticsearch

Testing: Pytest, Jest, Cypress, Selenium, k6, JMeter

Others: RESTful API, GraphQL, gRPC, WebSocket, Grafana, Prometheus, Datadog, New Relic, Git, GitHub, GitLab, Jira, Confluence, Agile, Scrum, Infrastructure as Code, HIPAA, PCI DSS

EXPERIENCE

Leanware May 2024–Present

Senior DevOps Engineer

- Led the redesign of cloud infrastructure for a healthcare SaaS platform across multiple microservices, implementing **Azure** and **Terraform** to establish Infrastructure as Code practices that enabled consistent, repeatable deployments across staging and production environments.
- Architected containerized workloads using **Docker** and **Azure Container Apps**, reducing deployment time from 2 hours to under 15 minutes and enabling the engineering team to ship critical healthcare features with confidence.
- Spearheaded the migration of CI/CD pipelines to **GitHub Actions**, automating healthcare compliance validation checks and reducing manual release reviews by 60%, allowing the team to deploy security-critical updates without weekend standby rotations.
- Implemented network security and access control using **Azure Key Vault** and **Network Security Groups**, meeting HIPAA-grade encryption and segregation requirements for patient data across development, staging, and production environments.

- Designed and deployed **Redis** caching layer for a care coordination service, reducing database query load by 45% during peak hours and improving API response times for clinical workflows from 1.2 seconds to 340 milliseconds.
- Collaborated with platform architects to integrate **API Gateway** and establish secure API contracts between healthcare services, enabling teams to deploy independently while maintaining strict audit compliance.
- Configured **Prometheus** and **Grafana** monitoring across health check endpoints, alerting services, and application logs, giving on-call engineers visibility into 12+ critical infrastructure components and reducing incident response time by 50%.
- Automated infrastructure provisioning using **Terraform** modules for stateless services, allowing new engineers to spin up isolated environments in under 10 minutes instead of manual 3-day setups.
- Conducted security hardening workshops covering **Azure** best practices, NSG configuration, and Key Vault rotation policies, bringing the team to enterprise-grade compliance standards required for multi-tenant healthcare customers.
- Maintained 99.95% uptime for production healthcare systems by proactively monitoring **CloudWatch** dashboards and triaging infrastructure incidents, fixing a recurring Azure Load Balancer state leak that had been causing 2–3 outages per quarter.

ThredUp Mar 2022–May 2024

Senior Backend Engineer

- Drove the infrastructure overhaul supporting a Databricks ML platform migration, designing **AWS** and **Kubernetes** clusters to handle 100K+ daily image processing jobs for the visual recognition pipeline that powered a 15% boost in sell-through rates.
- Implemented **Docker** containerization for 8+ microservices in the ML platform, standardizing build and deployment workflows across data science and backend teams, which cut environment setup time from 4 hours to 12 minutes.
- Built automated deployment pipelines using **GitHub Actions**, coordinating feature gates and canary rollouts for the AI Shopping search feature, enabling weekly releases without manual coordination across backend, ML, and mobile teams.
- Scaled **PostgreSQL** database infrastructure to support 172M+ item catalogs and 55K brands, partitioning tables by brand and implementing read replicas that reduced query latency from 800ms to 210ms during peak traffic.
- Deployed **Redis** distributed caching for personalization models, reducing redundant ML inference calls by 72% and cutting per-request latency for the single-SKU recommendation engine from 650ms to 180ms.
- Integrated **API Gateway** and **CloudFlare** to enforce rate limiting and DDoS protection across public-facing endpoints, handling 2.8% conversion increase from iOS mobile traffic without infrastructure incidents.
- Configured comprehensive monitoring using **Datadog** and **New Relic**, tracking 40+ key infrastructure metrics across containers, databases, and APIs, which enabled the team to detect and fix a memory leak in the image resizing service before production impact.

- Collaborated with ML engineers to operationalize the garment measurement model using **Kubernetes** and custom scaling policies, reducing model serving inference time from 2 quarters of iterative optimization to 1 month of production-ready performance tuning.
- Documented infrastructure patterns and operational runbooks, enabling 3 junior engineers to independently deploy services and respond to common alerts without escalation to on-call senior engineers.

Spring Health Nov 2020–Mar 2022

Backend Engineer

- Played a key role in scaling the precision-matching ML platform across **AWS** and **Kubernetes** infrastructure to support 6x growth to 2M+ members, ensuring clinical validity was preserved through every deployment across staging and production.
- Engineered the Care Navigation platform infrastructure using **Terraform** and **Docker**, establishing Infrastructure as Code practices that enabled the global family launch with dependent eligibility matching across 20+ jurisdictions without manual environment configuration.
- Implemented HIPAA-compliant infrastructure on **AWS** using **Azure Key Vault** patterns and encrypted **PostgreSQL** replicas, supporting Fortune 500 onboarding (PepsiCo, Instacart, Bain) and passing security audits within 2 weeks of deployment.
- Built automated CI/CD workflows using **GitHub Actions** for healthcare-grade deployments, integrating compliance validation checks and automated rollback triggers that prevented 3 critical incidents from reaching production.
- Deployed operational ML pipelines on **Kubernetes** for dropout risk prediction and demand forecasting models, integrating **Prometheus** metrics to track model accuracy and trigger alerts when prediction drift exceeded clinical thresholds.
- Configured multi-region **AWS** failover and disaster recovery procedures, testing recovery time objectives monthly and reducing failover time from 45 minutes to under 8 minutes using **CloudWatch** cross-region replication.
- Mentored 2 junior backend engineers on cloud infrastructure patterns, pair-programming on service deployments and incident response, resulting in both engineers leading independent on-call rotations within 6 months.
- Streamlined environment setup for the global platform by creating **Docker** compose templates and **Terraform** modules, reducing new engineer onboarding from 5 days to 1 day and accelerating feature development velocity by 25%.

Tencent Games Nov 2014–Nov 2020

Infrastructure Engineer

- Built and maintained cloud infrastructure on GCP and internal Tencent platforms supporting Honor of Kings, scaling from launch to 100M daily active users by managing **Kubernetes** clusters, **Docker** containers, and TcaplusDB distributed database systems handling 40M+ daily transactions.

- Implemented game server orchestration using custom **Kubernetes** controllers and **Helm** templates, automating the deployment of 500+ game servers across regions and reducing manual scaling operations from weekly to fully automated daily adjustments.
- Engineered the Tencent Gaming Buddy emulator infrastructure using **Docker** containerization and **Linux** optimization, supporting 5M+ concurrent users and achieving 99.8% uptime through custom health checks and automated failover policies.
- Deployed comprehensive monitoring and alerting using internal metrics platforms and **Prometheus**, tracking server health, player latency, and anti-cheat system triggers across 8 game titles, enabling on-call teams to detect and respond to incidents in under 5 minutes.
- Integrated the ACE anti-cheat kernel-level module into game server infrastructure, coordinating **Linux** kernel patches and security updates that blocked 12M+ unauthorized access attempts over 3 years without false positives.
- Managed cross-region network optimization and **API Gateway** configurations for Arena of Valor globalization, reducing player latency from 350ms to 85ms in European regions and improving matchmaking reliability to 99.2% connection stability.
- Automated infrastructure provisioning for EA FIFA Online 3 and FIFA Online 4 platforms using custom **Terraform**-like IaC tooling, enabling the publisher to scale server capacity on-demand during seasonal events handling 8M+ concurrent players.
- Collaborated with game development studios across Tencent to establish **Git** and **GitHub**-based deployment workflows, standardizing release procedures and reducing production hotfix deployment time from 6 hours to 30 minutes.

EDUCATION

The Hong Kong Polytechnic University Sep 2010–Nov 2014
Bachelor of Science in Computer Science